

House Price Prediction – Project Summary

Introduction

The objective of this project was to develop a machine learning model capable of predicting house prices based on various housing features. The dataset contained information such as area, number of bedrooms, bathrooms, stories, parking spaces, furnishing status, and other amenities. The project involved data preprocessing, model training, evaluation, and visualization to understand the factors that influence house prices.

Data Preprocessing

The dataset was loaded and explored using Pandas. Categorical variables such as main road access, guest room availability, basement availability, air conditioning, and furnishing status were converted into numerical format using one-hot encoding. Duplicate records were checked and removed when necessary to ensure data quality.

Model Development

The dataset was divided into training and testing sets using an 80:20 ratio. Two regression models were implemented and compared:

1. Linear Regression
2. Random Forest Regressor

Both models were trained using the training dataset and evaluated on the testing dataset.

Model Evaluation

The performance of the models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Linear Regression Results

- MAE: 970,043.40
- RMSE: 1,324,506.96
- R^2 Score: 0.653

Random Forest Regressor Results

- MAE: 1,021,546.04
- RMSE: 1,400,565.97
- R^2 Score: 0.612

The results indicate that Linear Regression performed better than Random Forest Regressor for this dataset, achieving a higher R^2 Score and lower prediction error.

Key Findings

Feature importance analysis revealed that bathrooms, air conditioning, hot water heating, preferred area, and furnishing status were the most influential factors affecting house prices. Houses with better amenities and facilities generally had higher market values.

Visualizations

Three visualizations were created to support the analysis:

1. Distribution of House Prices Histogram
2. Correlation Heatmap
3. Actual vs Predicted Price Scatter Plot

These visualizations helped in understanding data distribution, feature relationships, and model performance.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. Among the models tested, Linear Regression provided the best performance with an R^2 Score of 0.653. The analysis highlighted the importance of housing amenities and property characteristics in determining house prices. The developed model can assist buyers, sellers, and real estate businesses in making more informed decisions regarding property valuation.